Skylar Tian

h.tian@columbia.edu | New York, NY

EDUCATION

Columbia College, Columbia University **New York, NY**

B.A. in Mathematics & Computer Science

- **GPA:** 3.94/4.00 | **Awards:** Dean's List (All Semesters), Laidlaw Scholar (1 of 25 students at Columbia College)
- **Coursework:** Topology, Fourier Analysis, Machine Learning, Statistical Inference, Probability Theory, Linear Algebra
- **Pre-College:** USAMO Participant, AIME 11/15, USAPhO qualifier, Stanford Math Camp (SUMaC) | **SAT:** 1590/1600

RESEARCH AND PROFESSIONAL EXPERIENCE

Columbia Zuckerman Institute **New York, NY**

Undergraduate Researcher, Computational Neuroscience *Jun 2026 – Present*

- Selected as 1 of 10 students for a competitive research program, working under Computational Neuroscientist Liam Paninski

Selected Project Experience: **Neural Dynamics and Behavior Prediction**

- Benchmarked local field potential (LFP) representations from lfpack and autoencoder models against raw recordings to evaluate preservation of neural dynamics at high temporal resolution, achieving a 95% reduction in spatial dimension
- Modeled simultaneous recordings across brain regions using shared and region-specific latent components to investigate global and local neural dynamics across spatial scales, including HMM and Switching Linear Dynamical Systems
- Ongoing: compare continuous, discrete, and hierarchical brain-state models to characterize dimensionality and timescales, testing whether global states modulate regional inputs, local computations, or both; aiming to develop a dynamic brain atlas

XPulse **New York, NY**

Software Developer *May 2026 – Present*

- Engineered a weekly publishing pipeline on FastAPI, Supabase PostgreSQL, and Railway workers, integrating scheduling and concurrency controls to coordinate background tasks and support reliable and consistent episodes
- Regularly upgraded content selection and OpenAI speech generation to newer audio models and added Mutagen duration validation holding episodes to 15–20 minutes, reducing cross-episode variability and improving reliability
- Automated cover-art generation with Gemini reference-conditioned prompts (eight palettes, pose, gaze) and validated image integrity, human figure anatomy, framing, and provenance, with agent push notification for final review gating publication

Laidlaw Foundation **New York, NY**

Student Researcher, Applied Mathematics *May 2026 – Aug 2026*

- Built a Python/Google OR-Tools routing model for 528 NYC food pantries and community kitchens, integrating OSRM travel times, vehicle capacities, and service windows to compare five allocation policies incorporating food insecurity rate
- Quantified equity-reach tradeoff across five solves per policy: need-prioritized routing increased median coverage of high-need demand from 26.3% to 57.7% with marginal reach decrease; tested sensitivity through skip-penalty perturbations

E Fund Management **Hong Kong, China**

Quantitative Research Winter Analyst *Dec 2025 – Feb 2026*

- Replicated Carhart Four-Factor and Fama-French Five-Factor asset pricing models on Chinese A-share equities, adapting factor construction, portfolio formation, and regression frameworks to unique market structure and regulatory environment

Selected Project Experience: **Deviation Salience Trading Strategy**

- Developed a reproducible Python pipeline covering Chinese equities from 2021–2025, constructing industry-relative salience signals and double-sorted portfolios to test next-month momentum and reversal under equal and value weighting
- Evaluated seven portfolio specifications and applied Fama–MacBeth regressions with size, momentum, and volatility controls; used Newey–West inference and missing-return sensitivity tests to assess return patterns across weighting schemes

Jane Street **New York, NY**

Participant, WiSE Program *Jun 2025 – Jul 2025*

- Selected as 1 of 100 students to participate in an intensive training program on trading, market-making, and decision theory
- Built and analyzed cross-sector trading models in simulated markets using probability, expected value, and capital allocation

LEADERSHIP AND EXTRACURRICULARS

Columbia Alpha Partners **New York, NY**

Executive Board, Treasurer *Sep 2025 – Present*

- Led an investment pod's copper research, translating geopolitical supply disruptions and demand from AI data centers, electric vehicles, and grid infrastructure into a structural pricing model with 500 Monte Carlo simulations for estimation

Columbia Sailing Team **New York, NY**

Race Team Member, Skipper & Crew *Sep 2025 – Present*

- Competed for Columbia Club Sailing Team

ADDITIONAL INFORMATION

Skills: Python, Java, MATLAB, Microsoft Office (Excel, PowerPoint, Word), Git, 2D-Design, PCA, Monte Carlo simulation
Interests: Columbia Sailing Team, Classical Music (Harp, Operatic Singing, Flute, Guitar, Piano), Undergraduate Math Journal